Did I Leave the Garage Open?

One Real-World Problem, Many Engineering Lessons





MOMENTUM
Developer Conference
powered by The Circuit

About Me

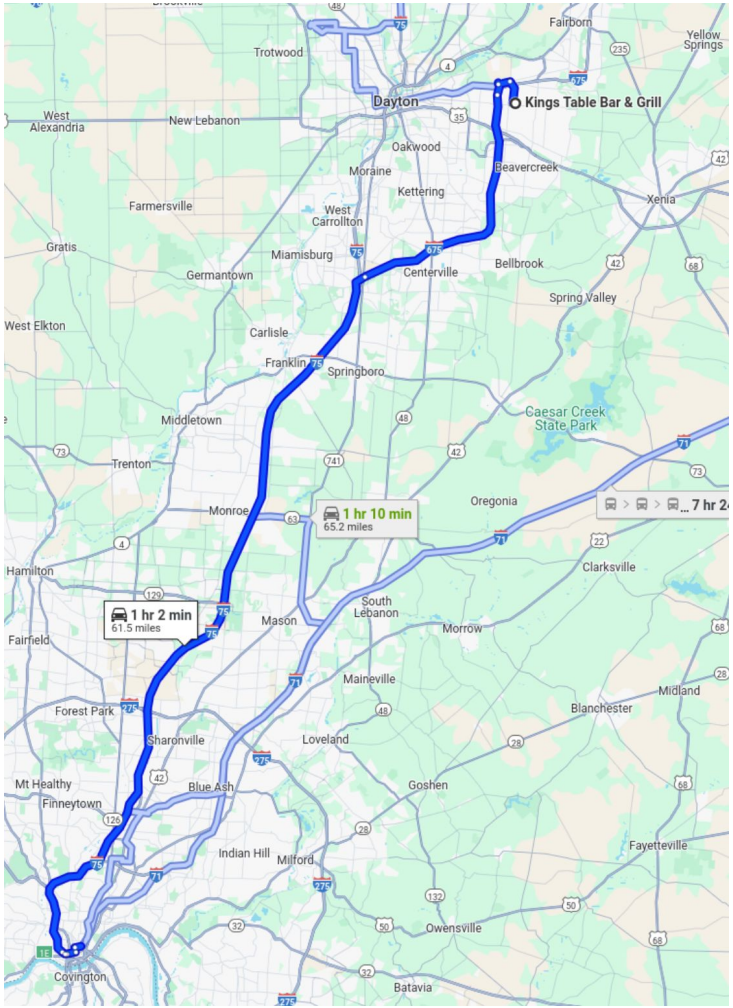
Live in Beavercreek with wife & two kids

Enjoy DIY projects, hiking, reading

Personal site: mattbanderson.com

Work for a cloud consulting company

APPLIED INFORMATION SCIENCES



Highlights

How to use Raspberry Pi & a magnetic door sensor to monitor whether a garage door is open or closed

Alerts based on open/closed state, duration, and time of day

Simulating a button press in software to open or close a garage door

Options for using cloud services for notifications

Engineering lessons learned along the way

Ideas for next steps and other integrations

DISCLAIMER

The information in this presentation is for educational, inspirational, and "wow-I-didn't-blow-anything-up" purposes only. Attempting to build your own Raspberry Pi garage door controller may result in newfound confidence, unexpected sparks (literal or metaphorical), or spontaneous Googling at 2 AM. I am not responsible for fried circuits, voided warranties, mildly judgmental looks from your electrician and/or significant other, or your decision to connect a toaster to the internet. Proceed with curiosity and caution, and maybe ask an electrician or electrical engineer first.


DIAMONDBACK
DIAMONDBACK

Internet # 322246530 Model # B2211T Store SKU # 1008433680

Top Rated

Chamberlain

1/2 HP Smart Quiet Belt Drive Garage Door Opener with Battery Backup

4.5 (1050) Questions & Answers (243)



Hover Image to Zoom

SUPER SAVINGS END

2 DAYS 17 HRS 13 MIN 20 SEC

$199.00

Pay **$174.00** after **$25 OFF** your total qualifying purchase upon opening a new card.

Apply for a Home Depot Consumer Card

- Ultra-quiet belt drive operation is perfect for attached garages
- Monitor and control your garage door with the myQ app
- Battery backup: open/close the garage door even when power is out
- View More Details

Beavercreek Store

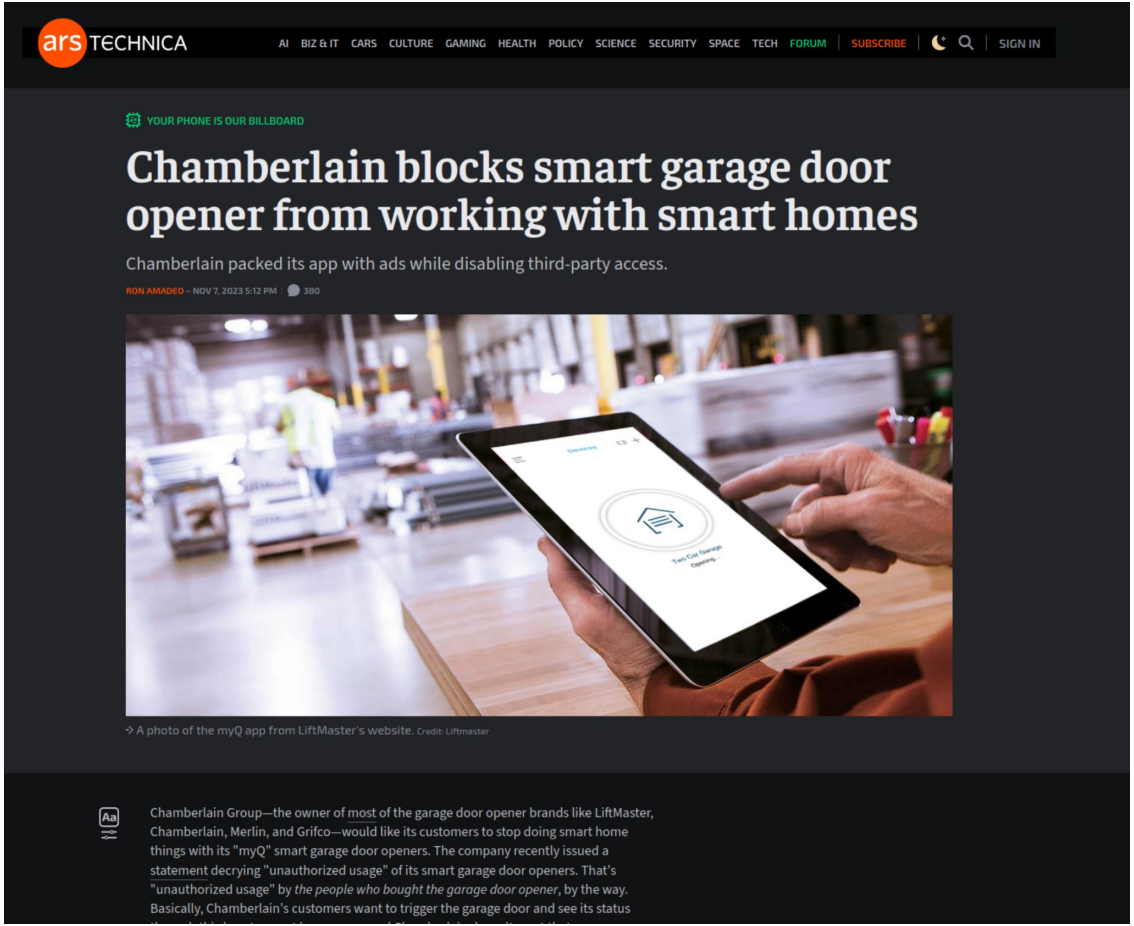
✓ 3 in stock Aisle 29, Bay 002

Pack Size: **1 Pack**

1 Pack | 2 Pack | 4 Pack

Pickup
Today
3 in stock
FREE

Delivery
Tomorrow
211 available
FREE



https://arstechnica.com/gadgets/2023/11/chamberlain-blocks-smart-garage-door-opener-from-working-with-smart-homes/



I Am Devloper

@iamdevloper

I’m a developer seeking a job in:

- a startup
- a corporate
- the woods, making furniture with my hands and not speaking to anyone


DO IT YOURSELF
FEEL A SENSE OF ACCOMPLISHMENT

Sensor Hardware Required

Raspberry Pi + power supply

SD card w/ OS installed

Female to female jumper wires

Magnetic sensor

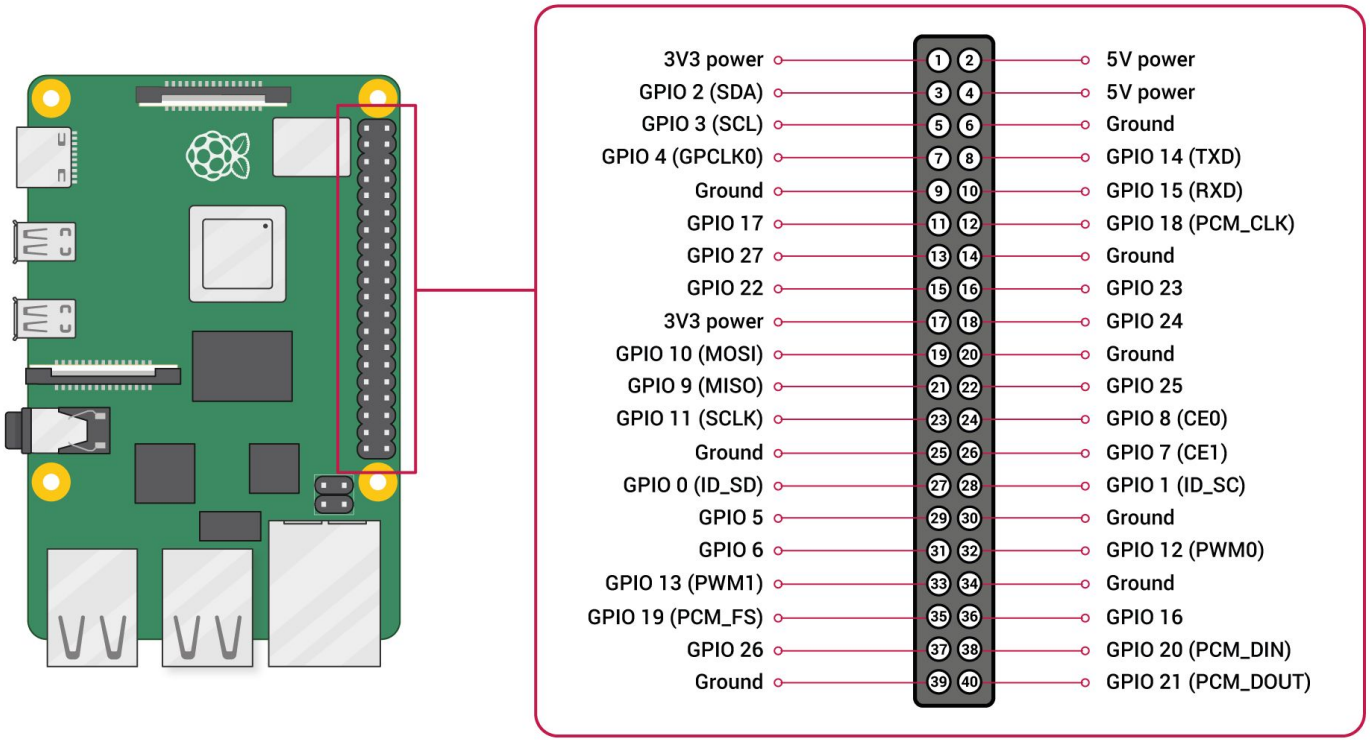
2-Conductor low-voltage wire (i.e. garage door/alarm/doorbell wire)

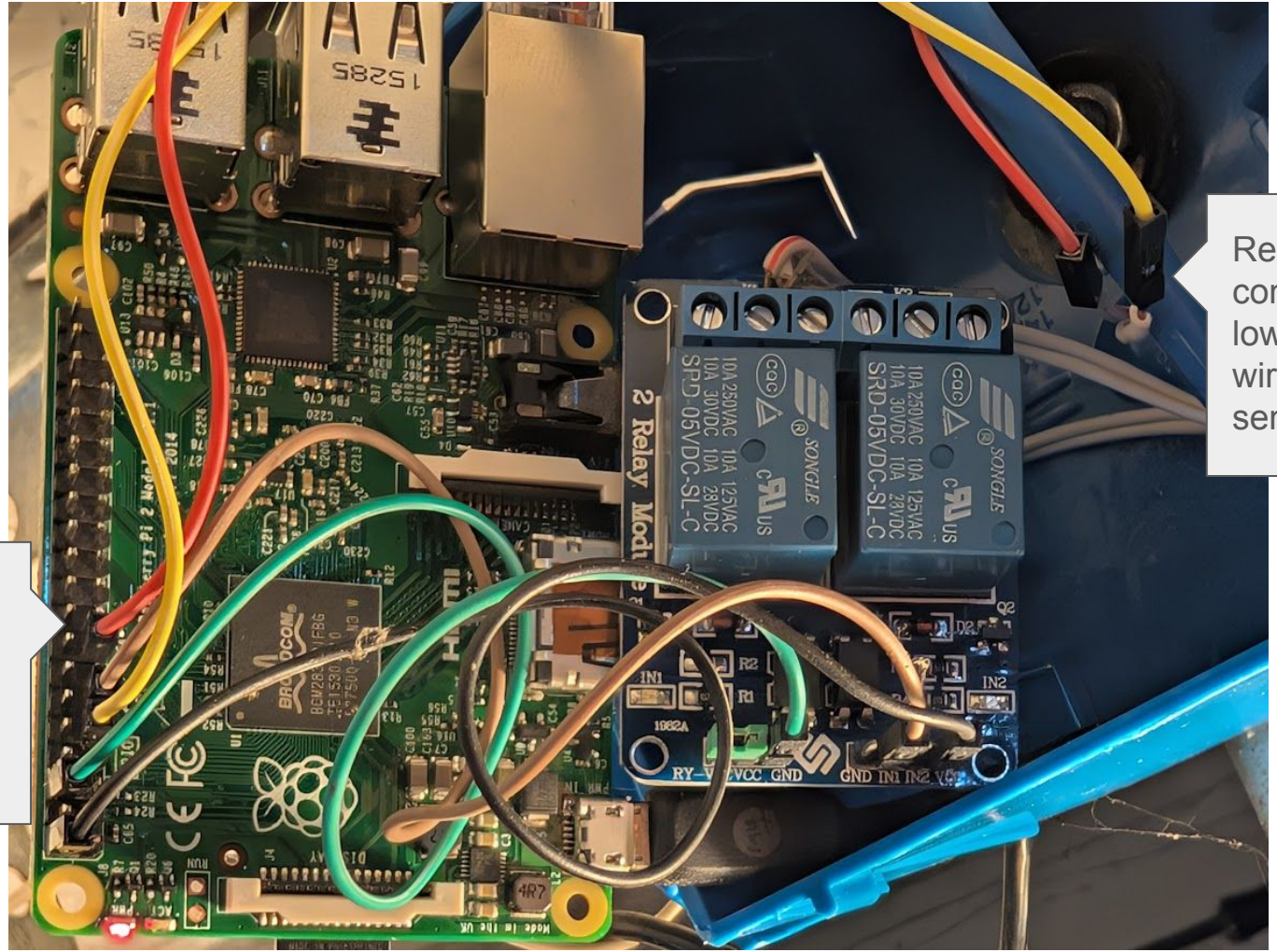
Sensor Setup

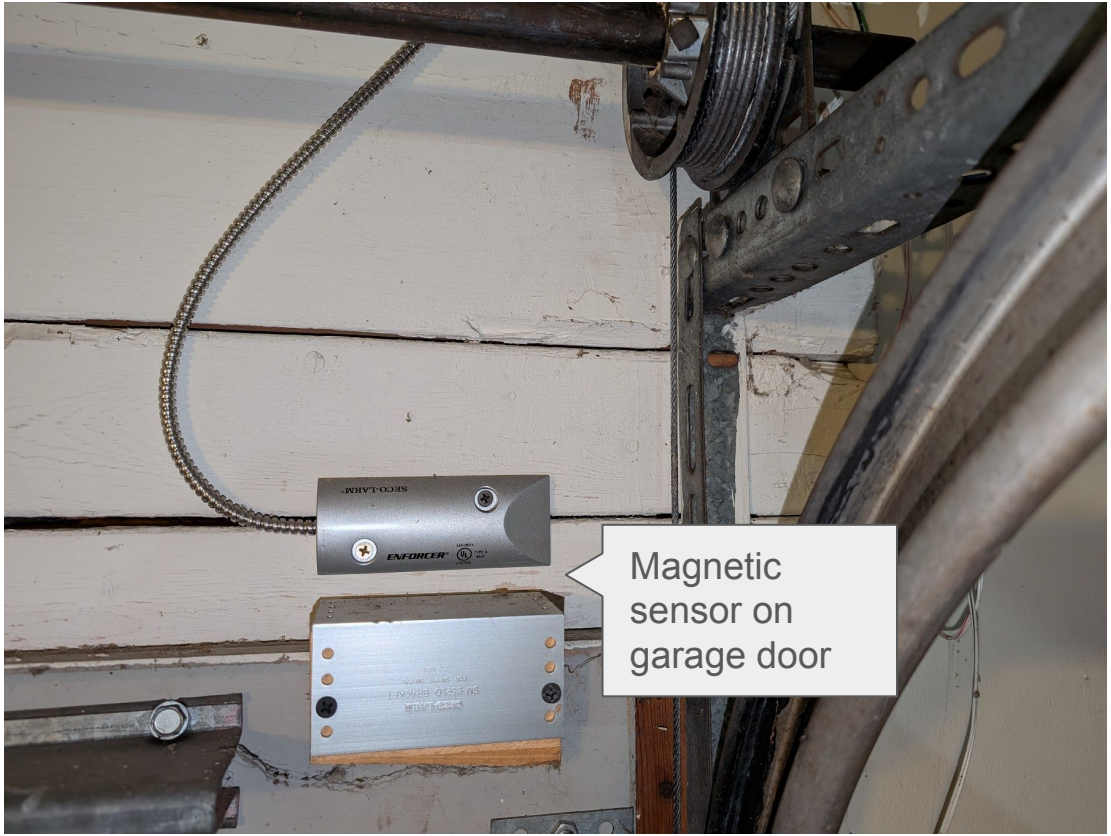
Set up Raspberry Pi with OS installed: https://www.raspberrypi.com/documentation/computers/getting-started.html#installing-the-operating-system

Connect magnetic sensor to RPi: Pin 9 (ground) & Pin 15 (GPIO 22)

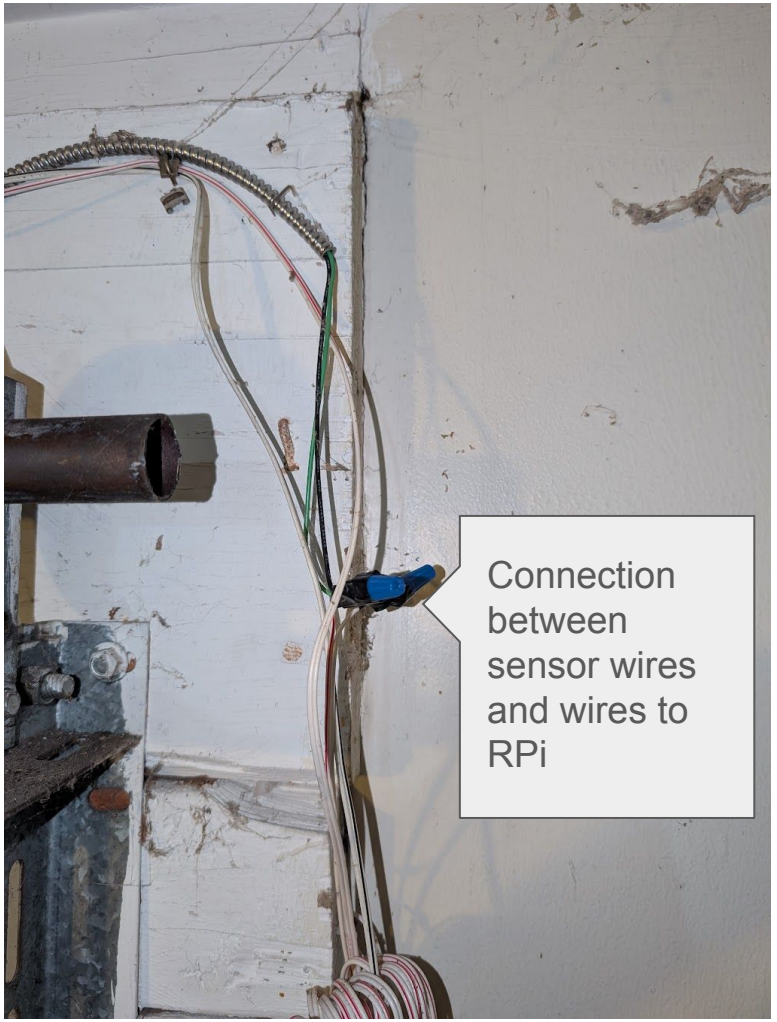
Raspberry Pi Pin Diagram




Red & yellow connect to low-voltage wire (to door sensor)
Red: Pin 15 (GPIO 22)
Yellow: Pin 9 (ground)
2 Relay Modu
SRD-05VDC-SL-C
SRD-05VDC-SL-C
SONGLE
10A 250VAC 10A 125VAC
10A 30VDC 10A 28VDC
IN1
IN2
VCC GND
GND IN1 IN2



Magnetic sensor on garage door



Connection between sensor wires and wires to RPi

Sensor Setup

#1 Recommendation:

Connect the door sensor and make sure it works before attaching it to the garage door…much easier to troubleshoot!

RPi App Configuration

/usr/local/etc/pi_garage_alert_config.py

```
GARAGE_DOORS = [{
        'pin': 15,
        'name': "Garage Door Name",
        'alerts': [
            {
                'state': 'open',
                'time': 600,
                'recipients': [ 'email:user@example.com', 'email:user2@example.com' ]
            }
        ]
    }]


LOC_LATITUDE = 40.7128

LOC_LONGITUDE = -74.0060
```

Location + Time of Day

/usr/local/etc/pi_garage_alert.py

```
city_loc_info = LocationInfo(city, country, timezone_str, cfg.LOC_LATITUDE, cfg.LOC_LONGITUDE)

local_dusk = sun(city_loc_info.observer, date=local_now)['dusk']

after_dusk = local_now > local_dusk

...

if time_in_state > alert['time'] and state == alert['state'] and after_dusk:

      send_alerts(...)
```

RPi App Setup

```
[sudo] apt-get install git python3-dev python3-pip
libopenblas-dev python3-rpi.gpio

git clone
https://github.com/mattbanderson/pi_garage_alert.git

cd pi_garage_alert

[sudo] mkdir /usr/local/sbin/pi_garage_alert

[sudo] cp bin/pi_garage_alert.py
/usr/local/sbin/pi_garage_alert

[sudo] cp requirements.txt
/usr/local/sbin/pi_garage_alert

[sudo] cp init.d/pi_garage_alert /etc/init.d/

[sudo] cp etc/pi_garage_alert_config.py
/usr/local/etc/
```

```
[sudo] chown pi
/usr/local/etc/pi_garage_alert_config.py

[sudo] chown pi -R
/usr/local/sbin/pi_garage_alert

cd /usr/local/sbin/pi_garage_alert

[sudo] python3 -m venv venv

source venv/bin/activate

pip3 install -r requirements.txt
```

Expected Output

```
[sudo] /usr/local/sbin/pi_garage_alert/pi_garage_alert.py

2024-05-12 17:13:16: ============================================================

2024-05-12 17:13:16: Pi Garage Alert starting

2024-05-12 17:13:16: Configuring global settings

2024-05-12 17:13:16: Configuring pin 15 for "Example Garage Door"

2024-05-12 17:13:16: Initial state of "Example Garage Door" is open

2024-05-12 17:13:20: CPU temp: 46.5, GPU temp: 46.0, Uptime: 0:50:17, Example Garage Door: open/0/4
```

Expected Output

```
[sudo] /usr/local/sbin/pi_garage_alert.py

2024-05-12 17:13:16: ============================================================

2024-05-12 17:13:16: Pi Garage Alert starting

2024-05-12 17:13:16: Configuring global settings

2024-05-12 17:13:16: Configuring pin 15 for "Example Garage Door"

2024-05-12 17:13:16: Initial state of "Example Garage Door" is open

2024-05-12 17:13:20: CPU temp: 46.5, GPU temp: 46.0, Uptime: 0:50:17, Example Garage Door: open/0/4

...

2024-05-12 17:13:30: State of "Example Garage Door" changed to closed after XXX sec
```

Email Notifications: Install & Configure Postfix

```
[sudo] apt-get install postfix mailutils libsasl2-2 ca-certificates libsasl2-modules

nano /etc/postfix/main.cf

nano /etc/postfix/sasl_passwd
```

Postfix Configuration

/etc/postfix/main.cf

```
relayhost = [smtp.example.com]:587

smtp_sasl_auth_enable = yes

smtp_sasl_password_maps = hash:/etc/postfix/sasl_passwd

smtp_sasl_security_options = noanonymous

smtp_tls_CAfile = /etc/ssl/certs/ca-certificates.crt

smtp_use_tls = yes
```

/etc/postfix/sasl_passwd

```
[smtp.example.com]:587 user@example.com:<password>
```

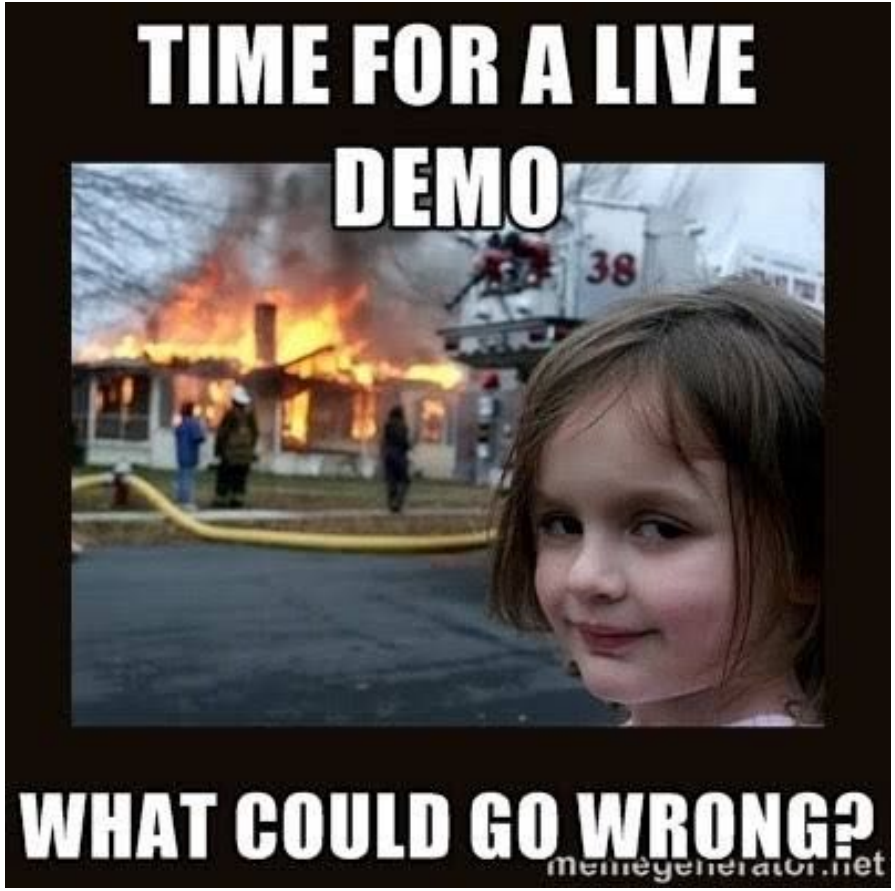
Apply Postfix Configuration

```
[sudo] chmod 400 /etc/postfix/sasl_passwd

[sudo] postmap /etc/postfix/sasl_passwd

[sudo] /etc/init.d/postfix reload

echo "test mail" | mail -s "test subject" recipient@example.com
```


TIME FOR A LIVE DEMO
38
WHAT COULD GO WRONG?

SMS Shortcut

Possible to send email as SMS using SMS gateways

https://en.wikipedia.org/wiki/SMS_gateway

Generally good enough for personal use

Mobile carrier	SMS gateway domain
Alltel[1]	sms.alltelwireless.com
AT&T[2]	txt.att.net
Boost Mobile[1]	sms.myboostmobile.com
Consumer Cellular[3]	mailmymobile.net
Cricket Wireless	mms.cricketwireless.net
FirstNet[5]	sms.firstnet.com 1waysms.firstnet.com
Google Fi Wireless[6]	msg.fi.google.com
MetroPCS	mymetropcs.com
Republic Wireless[7]	text.republicwireless.com
Sprint[1]	messaging.sprintpcs.com
T-Mobile[1]	tmomail.net
Ting[3]	message.ting.com
U.S. Cellular[1]	email.uscc.net
Verizon Wireless[8]	vtext.com
Virgin Mobile[1]	vmobl.com
XFinity Mobile[3]	vtext.com

Config with Email & “SMS”

/usr/local/etc/pi_garage_alert_config.py

```
GARAGE_DOORS = [{
        'pin': 15,
        'name': "Garage Door Name",
        'alerts': [
            {
                'state': 'open',
                'time': 600,
                'recipients': [ 'email:user@example.com', 'email:19998675309@vtext.com' ]
            }
        ]
    }]
```

Sends an email, arrives as SMS


Talk to me, talk to me, talk to me, talk to me

Hardware Required

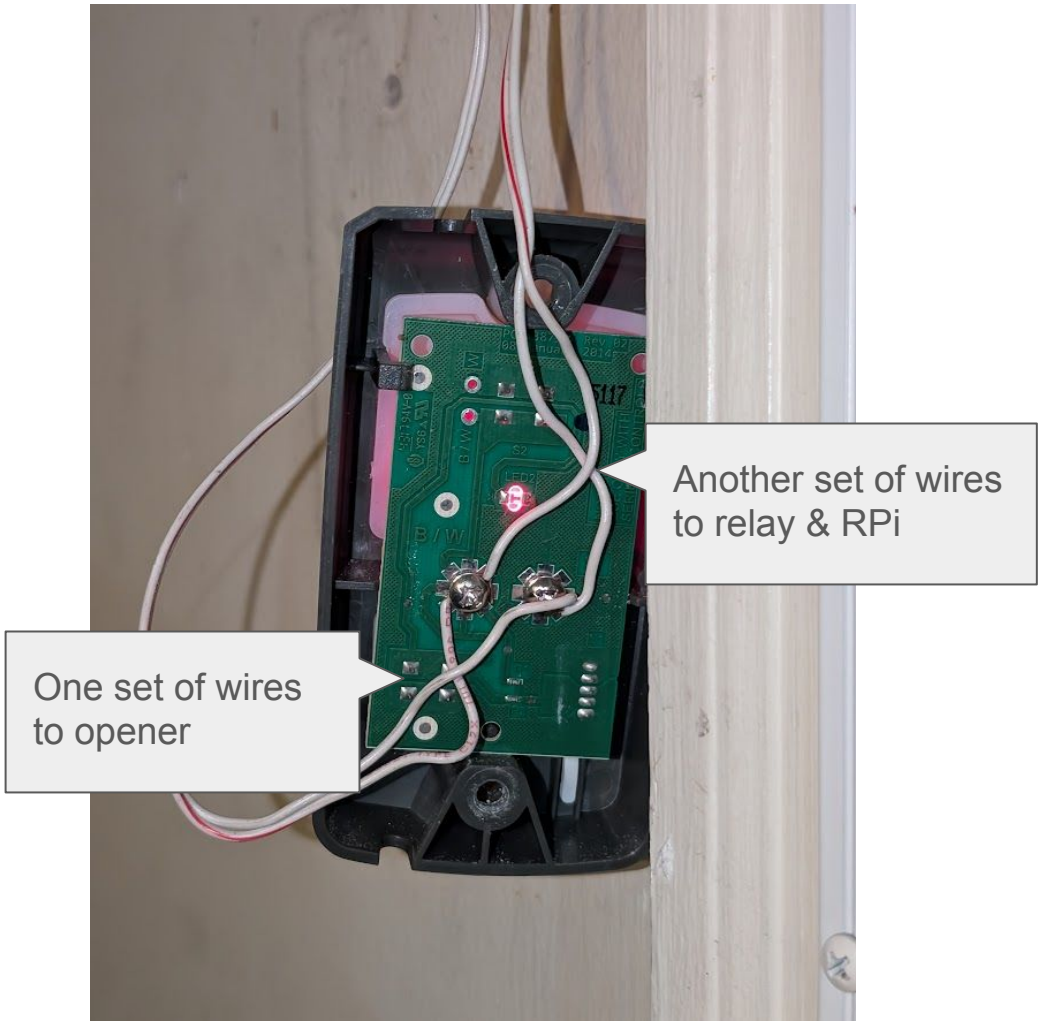
~~Raspberry Pi + power supply~~

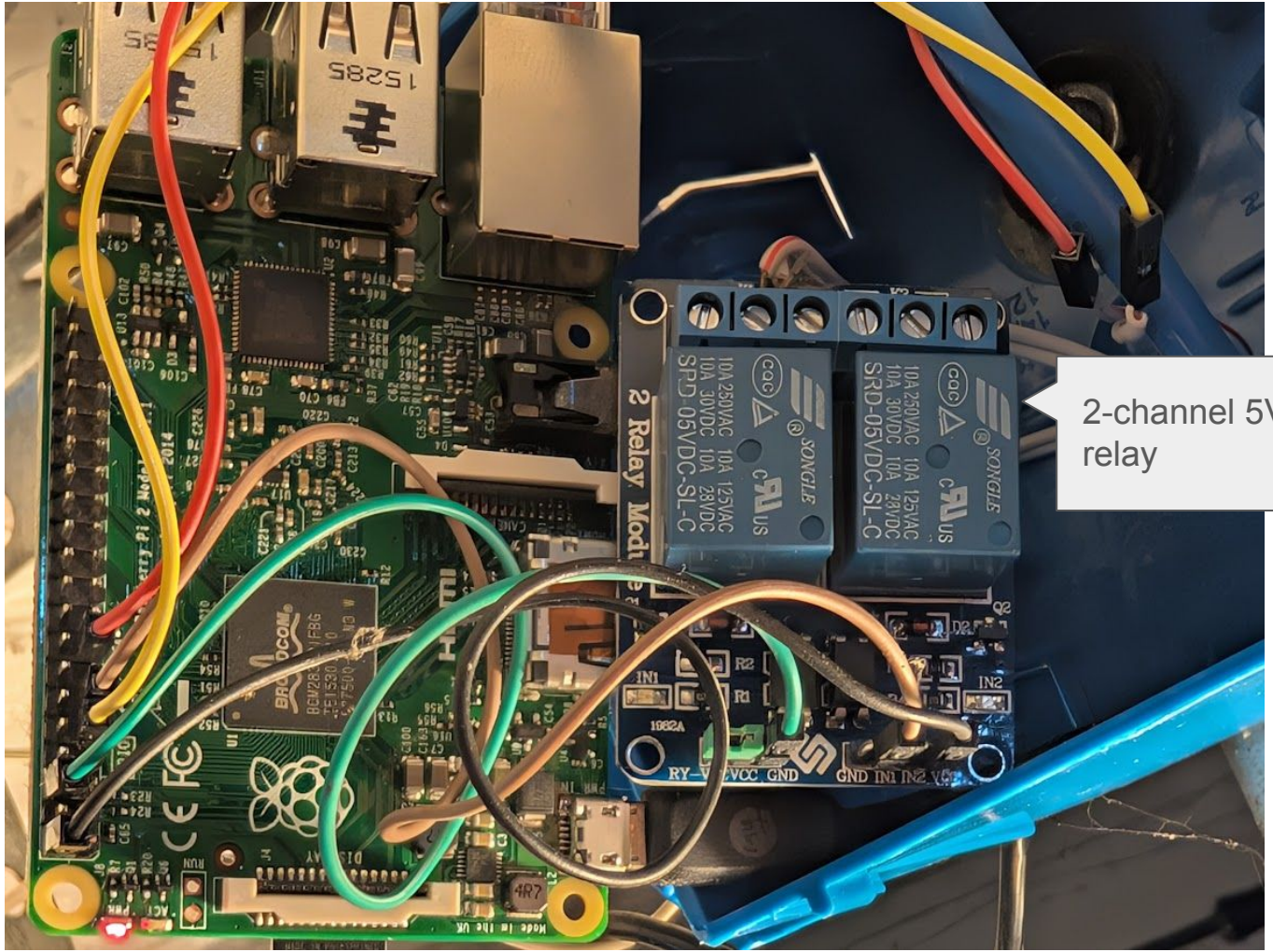
~~SD card w/ OS installed~~

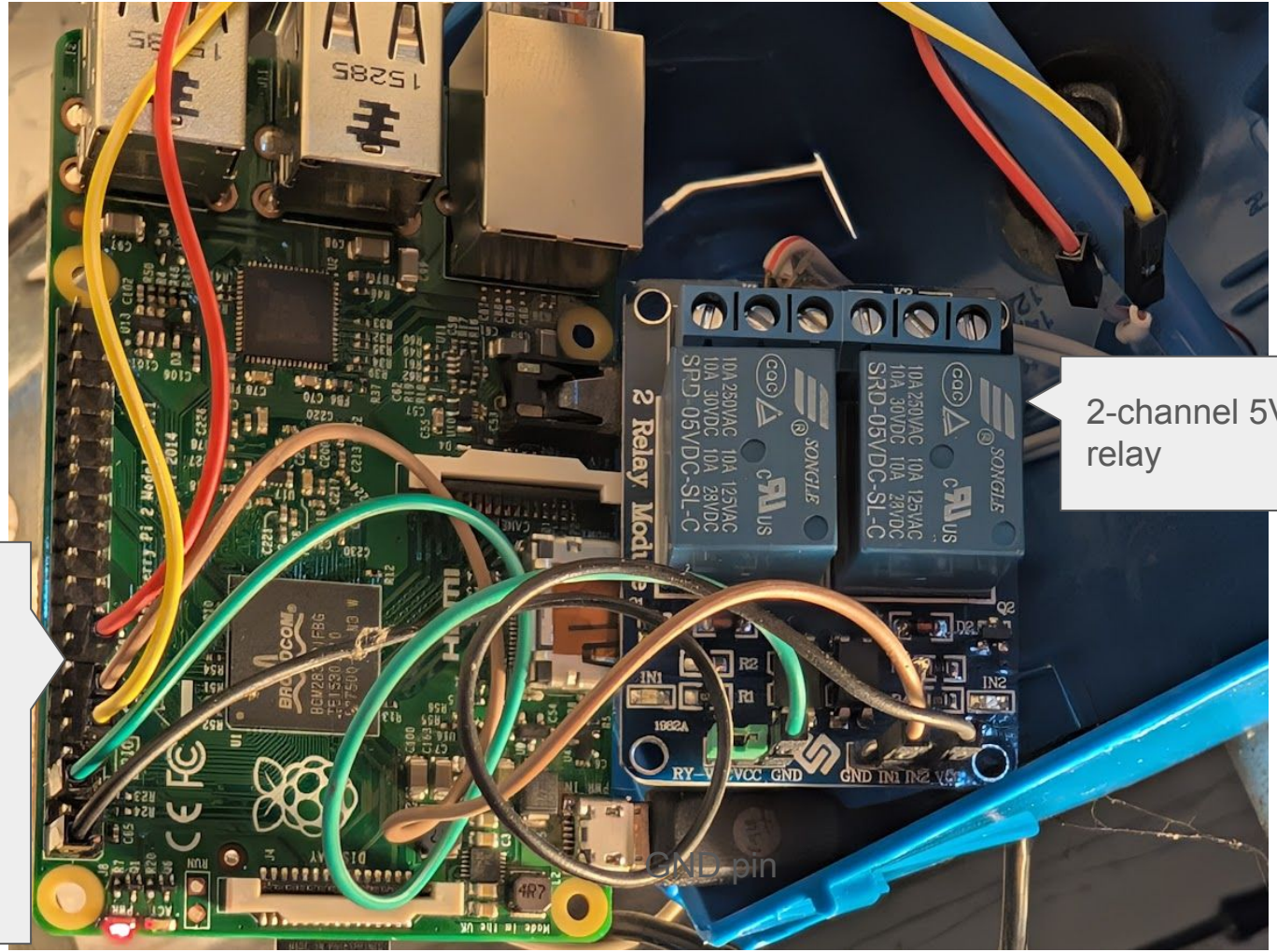
~~Female to female jumper wires~~

~~2 Conductor wire (i.e. garage door wire)~~

2 channel 5V relay


Another set of wires
to relay & RPi
One set of wires
to opener


2-channel 5V relay

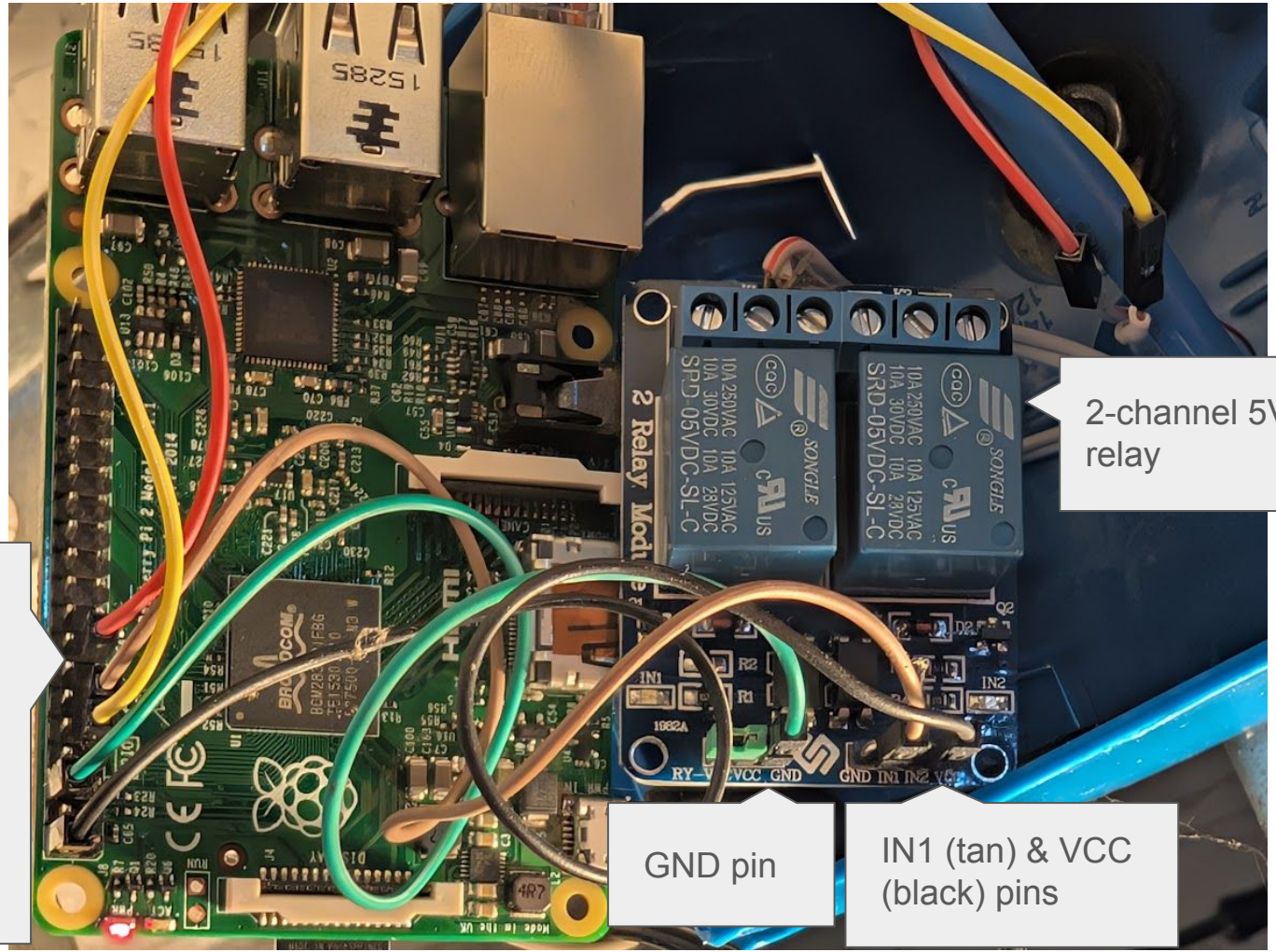


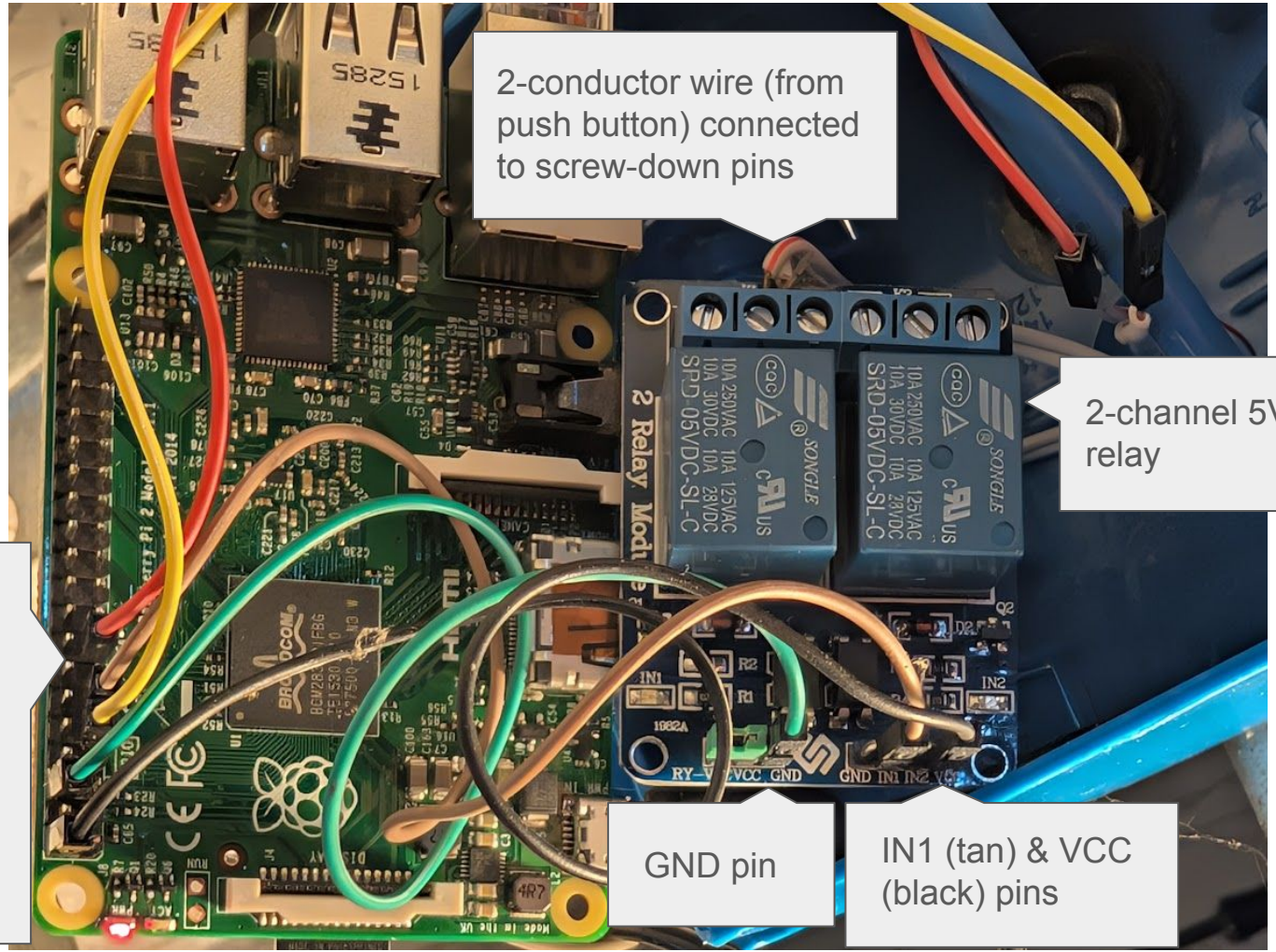
2-channel 5V relay

Black: Pin 2 (5V power)

Green: Pin 6 (ground)

Tan: Pin 11 (GPIO 17)


2-channel 5V relay
Black: Pin 2 (5V power)
Green: Pin 6 (ground)
Tan: Pin 11 (GPIO 17)
GND pin
IN1 (tan) & VCC (black) pins


2-conductor wire (from push button) connected to screw-down pins
2-channel 5V relay
Black: Pin 2 (5V power)
Green: Pin 6 (ground)
Tan: Pin 11 (GPIO 17)
GND pin
IN1 (tan) & VCC (black) pins

Software for Interacting with the RPi

```
[sudo] apt-get install nodejs npm

[sudo] npm install forever -g

git clone https://github.com/mattbanderson/garage-node

cd garage-node

nano config.js
```

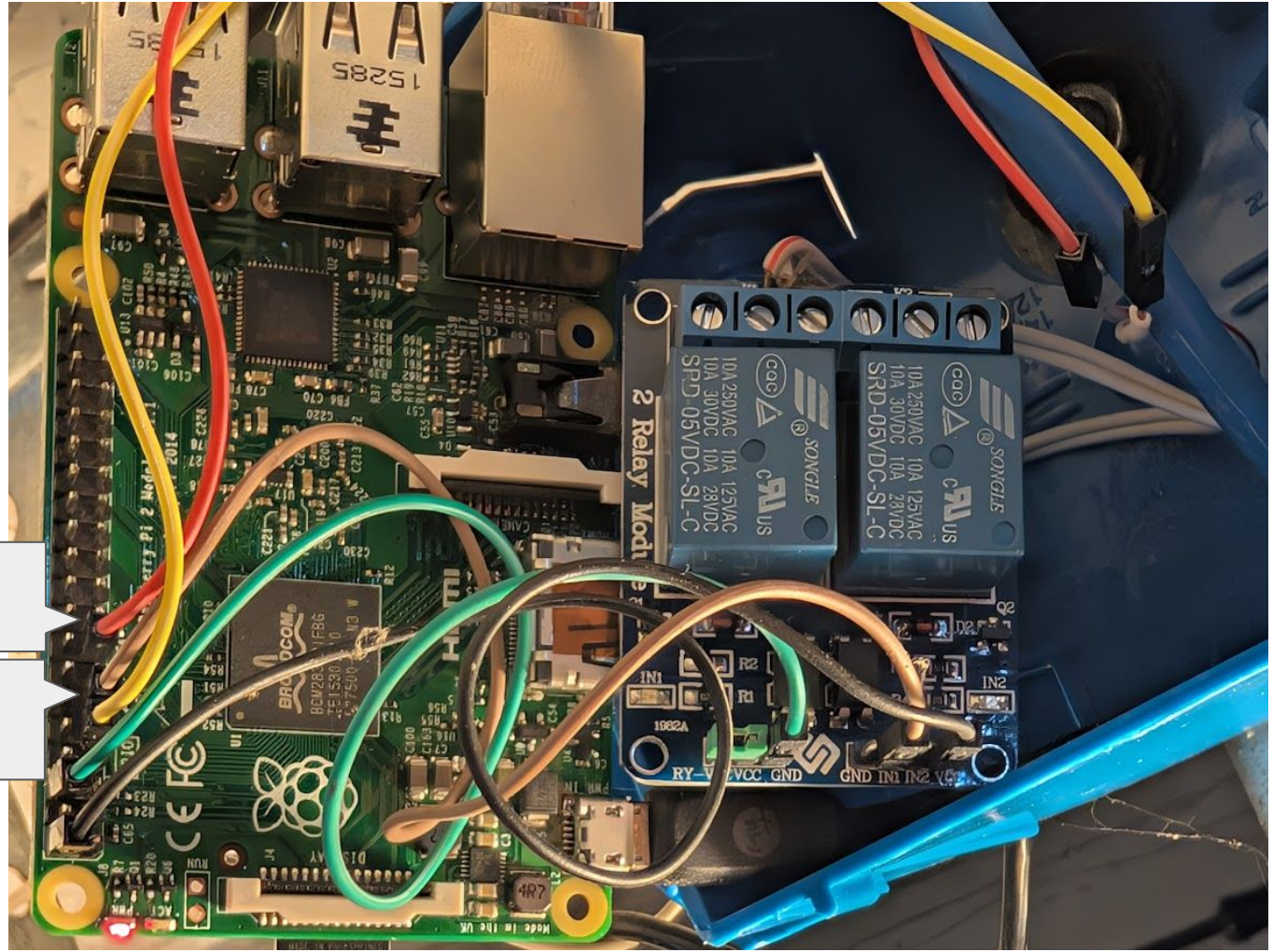
Node App Configuration

```
config.GARAGE_DOORS = [
  {
    name: 'Garage Door',
    writePin: 11,
    readPin: 15
  },
  {
    name: 'Garage Door #2',
    writePin: 16,
    readPin: 22
  }
];
config.RELAY_TIMEOUT = 500;
```

Node App Configuration

```
config.GARAGE_DOORS = [
  {
    name: 'Garage Door',
    writePin: 11,
    readPin: 15
  },
  {
    name: 'Garage Door #2',
    writePin: 16,
    readPin: 22
  }
];
config.RELAY_TIMEOUT = 500;
```




Red: Pin 15
(GPIO 22)
Tan: Pin 11
(GPIO 17)

Testing

```
npm install

npm start

curl http://localhost:3000/api/ping

     Returns “pong”

curl -X POST http://localhost:3000/api/garage/door/1

     Should open or close the garage door
```

Always-on Setup

Using Forever:

```
forever start -o /home/pi/garage-node/out.log -e /home/pi/garage-node/err.log
/home/pi/garage-node/app.js
```

As a service:

```
/etc/systemd/system/garage-node.service  # create file (details in repo)
[sudo] systemctl daemon-reload
[sudo] systemctl enable garage-node.service  # enable/start on boot
[sudo] systemctl start garage-node.service   # start now
[sudo] systemctl status garage-node.service  # verify running
[sudo] journalctl -u garage-node.service -f  # another way to verify
```

Configuring Other Alerts

Twilio, Slack, X/Twitter, etc,

pi_garage_alert.py:

```
class Twilio:
...
class Slack:
...


def send_alerts(...):

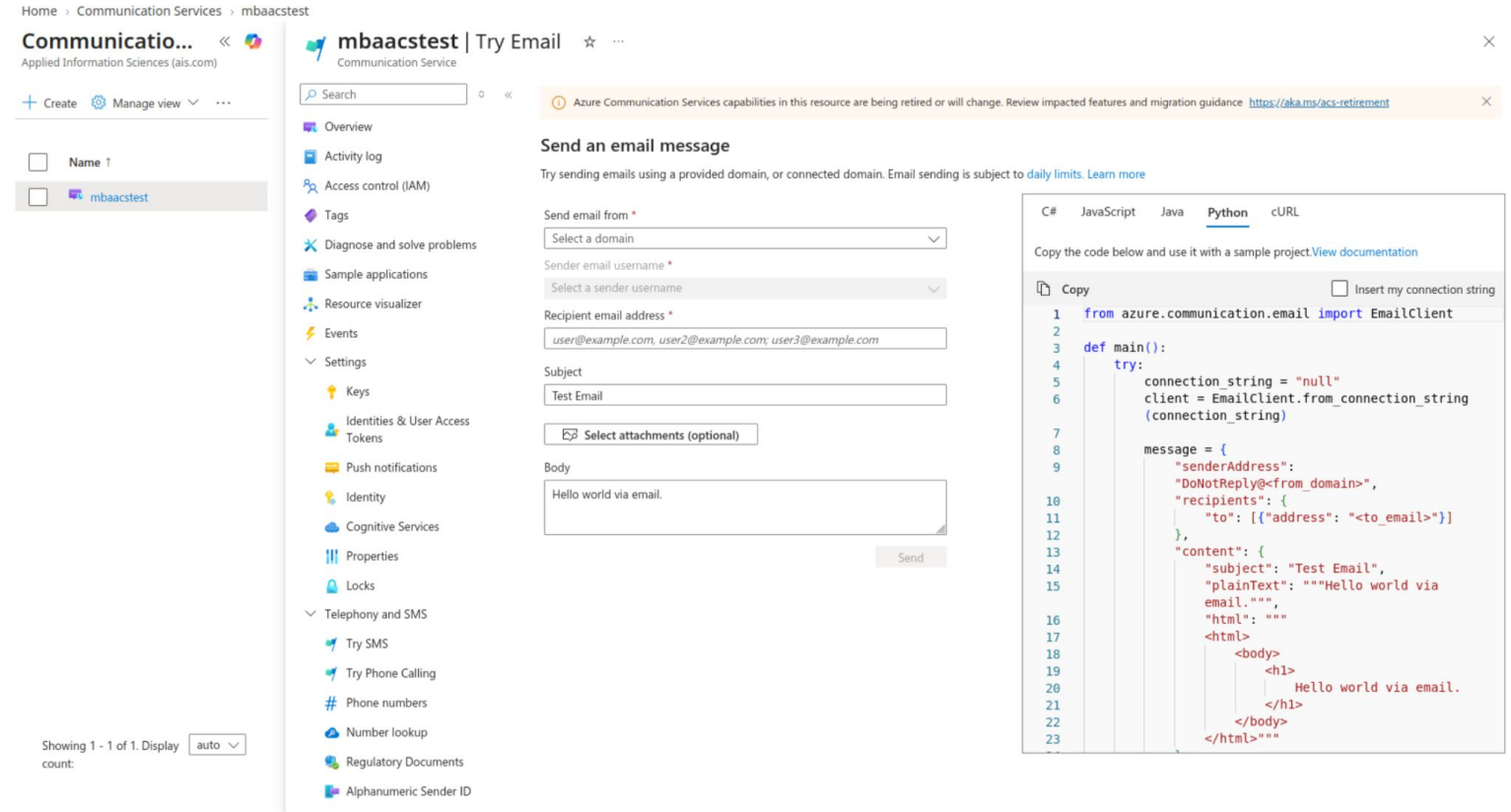
     # mapping of alert type keys to services
```

Configuring Other Alerts

pi_garage_alert_config.py:

```
GARAGE_DOORS = [{
        'pin': 15,
        'name': "Garage Door Name",
        'alerts': [
            {
                'state': 'open',
                'time': 600,
                'recipients': [
                      'email:user1@example.com',
                      'email:user2@example.com',
                      'sms:19998675309',
                      'acs_email:user@example.com' ]
            }
        ]
    }]
```

Add a New Alert Service



Add ACS Email support

```
class AzureAcsEmail:

    def send_email(self, recipient, subject, msg):
        self.logger.info("Sending email to %s: subject = \"%s\", message = \"%s\"", recipient, subject, msg)

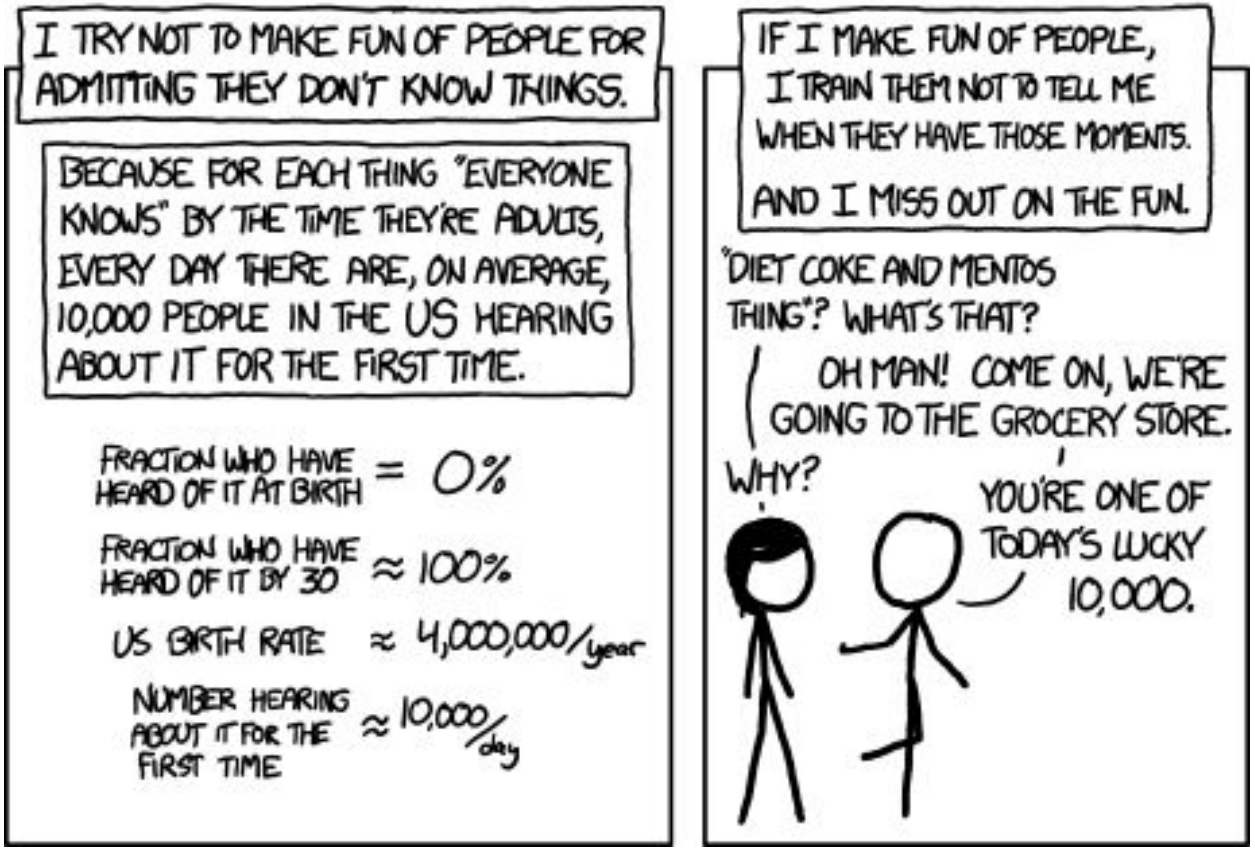
        try:
            connection_string = cfg.ACS_CONN_STR
            client = EmailClient.from_connection_string(connection_string)

            message = {
                "senderAddress": cfg.ACS_FROM,
                "recipients": {
                    "to": [{"address": recipient}]
                },
                "content": {
                    "subject": subject,
                    "plainText": msg
                },

            }

            poller = client.begin_send(message)
            result = poller.result()
            self.logger.info("ACS email message sent")

        except Exception as ex:
            self.logger.exception(ex)
```


I TRY NOT TO MAKE FUN OF PEOPLE FOR ADMITTING THEY DON'T KNOW THINGS.
BECAUSE FOR EACH THING "EVERYONE KNOWS" BY THE TIME THEY'RE ADULTS, EVERY DAY THERE ARE, ON AVERAGE, 10,000 PEOPLE IN THE US HEARING ABOUT IT FOR THE FIRST TIME.
FRACTION WHO HAVE HEARD OF IT AT BIRTH = 0%
FRACTION WHO HAVE HEARD OF IT BY 30 ≈ 100%
US BIRTH RATE ≈ 4,000,000/year
NUMBER HEARING ABOUT IT FOR THE FIRST TIME ≈ 10,000/day
IF I MAKE FUN OF PEOPLE, I TRAIN THEM NOT TO TELL ME WHEN THEY HAVE THOSE MOMENTS.
AND I MISS OUT ON THE FUN.
"DIET COKE AND MENTOS THING"? WHAT'S THAT?
OH MAN! COME ON, WE'RE GOING TO THE GROCERY STORE.
WHY?
YOU'RE ONE OF TODAY'S LUCKY 10,000.

Engineering Lesson #1

Reliability is far and away the most important feature

- Thoroughly explain physical/manual backup options
- Test, test, test!

Engineering Lesson #2

AI tools dramatically shorten the learning curve

- Explaining how concepts work
- Retaining context so you don’t have to

Engineering Lesson #3

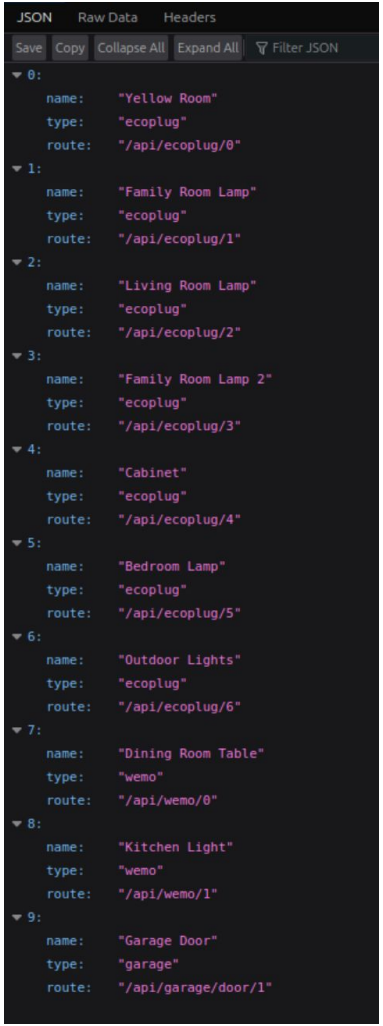
3rd party tools/integrations vs roll-your-own

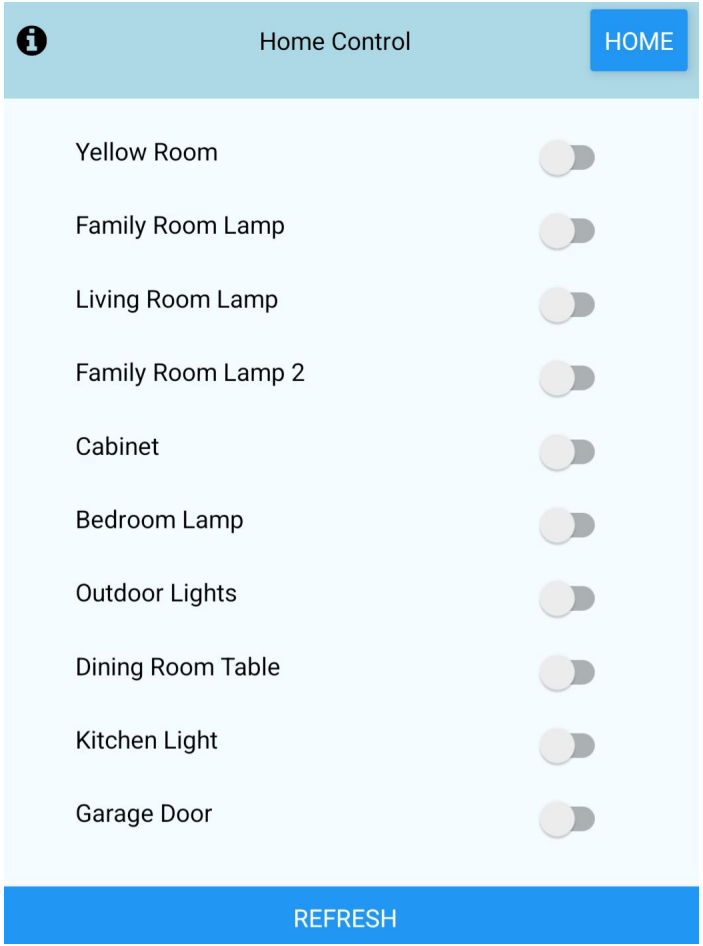
- What do you want to spend your time on?
- Value of open source software

What's Next?

Create a custom REST API

Custom app frontend or mobile app





What's Next?

Integrate with Home Assistant



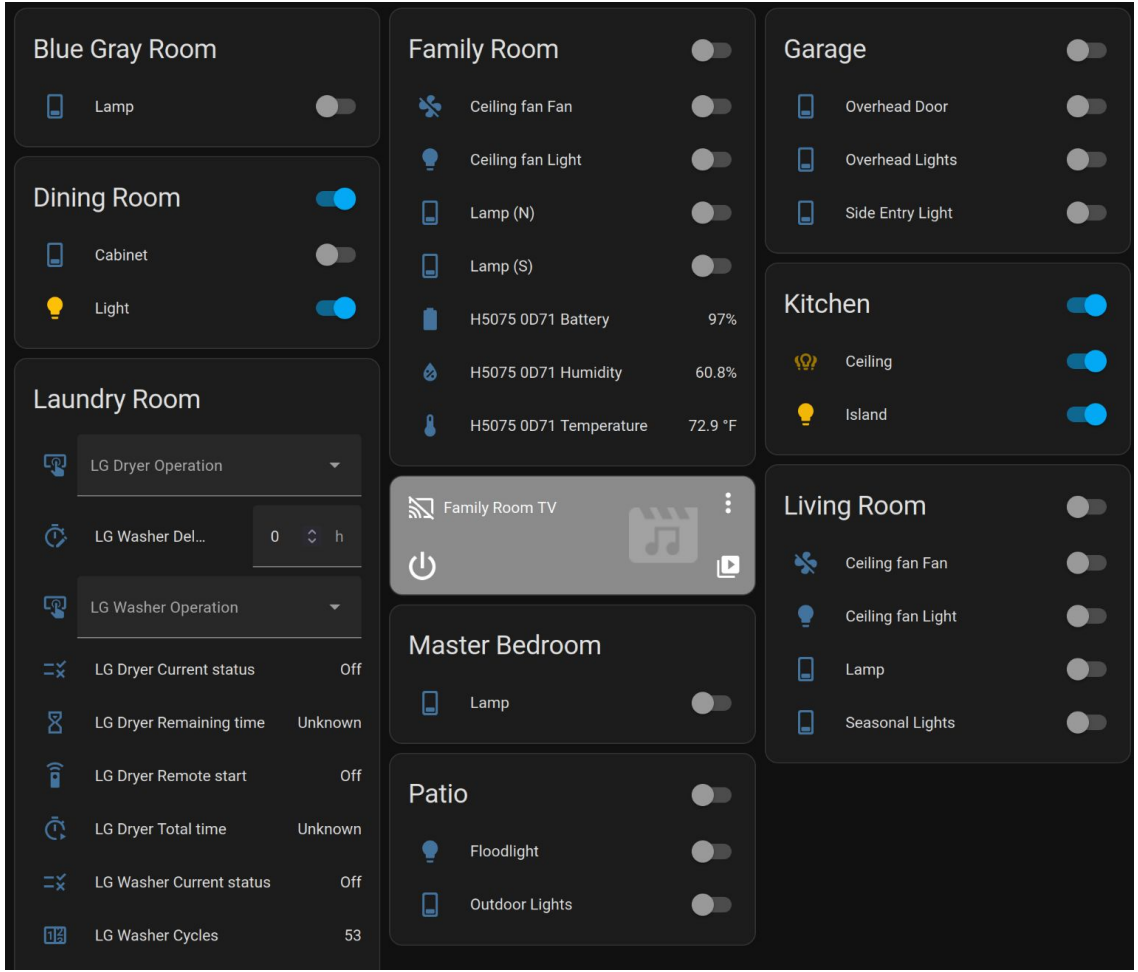
/homeassistant/configuration.yaml

```
# Loads default set of integrations. Do not remove.
default_config:

# Load frontend themes from the themes folder
frontend:
  themes: !include_dir_merge_named themes

automation: !include automations.yaml
script: !include scripts.yaml
scene: !include scenes.yaml

switch:
  - platform: rest
    name: 'Garage Overhead Door'
    unique_id: garage_overhead_door
    resource: http://xxx.xxx.x.xxx:8080/api/garage/door/1
    is_on_template: '{{ value_json == 1 }}'
```



Garage Pi Tips

Use a good power supply

Easy to mix up RPi Pin number and GPIO number

Make sure SSH is enabled

Decide on Ethernet vs WiFi

No idea if anything besides Email/SMS works

Rapid requests can cause issues

Send test alerts to verify communication status

Closing Thoughts

Code might be a bit outdated, 10+ years old

Should work on any type of on/off or open/closed sensor

If it breaks/malfunctions, it’s your fault

Maintenance & upgrades

But it just works!

Everything is local!

Other Options

OpenGarage: $50, open source, https://opengarage.io/

Other projects: https://hn.algolia.com/?q=garage+door

Parts list

Raspberry Pi: https://link.amazon/B03tJoFAX (~$55) Kit: https://link.amazon/B02rJs1KV (~$90)

Power supply: https://link.amazon/B0f9dpTsq (~$10)

SD card

Raspberry Pi OS on SD card: https://www.raspberrypi.com/software/

2 channel 5V relay: https://link.amazon/B023HlQ6S (~$9)

Female to female jumper wires: https://link.amazon/B04pXpzXP (~$7)

Magnetic garage door sensor: https://link.amazon/B03NCgxF5 (~$20)

2 Conductor wire (i.e. garage door wire): https://link.amazon/B0frLThnM (~$7)

Links

Raspberry Pi setup, part 1:
https://web.archive.org/web/20230406173732/https://www.richlynch.com/2013/07/27/pi_garage_alert_1/

Raspberry Pi setup, part 2:
https://web.archive.org/web/20230311070645/https://www.richlynch.com/2013/08/04/pi_garage_alert_2/

Raspberry Pi sensor reading and alerts:

- Mine: https://github.com/mattbanderson/pi_garage_alert
- Original: https://github.com/rllynch/pi_garage_alert

Raspberry Pi button press simulation:

- Mine: https://github.com/mattbanderson/garage-node
- Original: https://github.com/brentnycum/garage-node

DISCLAIMER

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